

```
In [ ]: import pandas as pd
```

```
In [18]: file_path = 'apartments_for_rent_classified_10K.csv' # Replace with the actual path to your ARFF file  
df = pd.read_csv('apartments_for_rent_classified_10K.csv', sep=';', encoding='cp1252', on_bad_lines='skip')
```

```
In [22]: df_1 = pd.read_csv('apartments_for_rent_classified_100K.csv', sep=';', encoding='cp1252', on_bad_lines='skip')
```

C:\Users\LUXAN\AppData\Local\Temp\ipykernel_8280\4132816156.py:1: DtypeWarning: Columns (15) have mixed types. Specify dtype option on import or set low_memory=False.

```
df_1 = pd.read_csv('apartments_for_rent_classified_100K.csv', sep=';', encoding='cp1252', on_bad_lines='skip')
```

```
In [19]: df.head()
```

Out[19]:

	id	category	title	body	amenities	bathrooms	bedrooms	currency	fee	has_photo	...	price_dis
0	5668626895	housing/rent/apartment	Studio apartment 2nd St NE, Uhland Terrace NE,...	This unit is located at second St NE, Uhland T...	NaN	NaN	0.0	USD	No	Thumbnail	...	\$
1	5664597177	housing/rent/apartment	Studio apartment 814 Schutte Road	This unit is located at 814 Schutte Road, Evan...	NaN	NaN	1.0	USD	No	Thumbnail	...	\$
2	5668626833	housing/rent/apartment	Studio apartment N Scott St, 14th St N, Arling...	This unit is located at N Scott St, 14th St N,...	NaN	1.0	0.0	USD	No	Thumbnail	...	\$1,
3	5659918074	housing/rent/apartment	Studio apartment 1717 12th Ave	This unit is located at 1717 12th Ave, Seattle...	NaN	1.0	0.0	USD	No	Thumbnail	...	\$
4	5668626759	housing/rent/apartment	Studio apartment Washington Blvd, N Cleveland ...	This unit is located at Washington Blvd, N Cle...	NaN	NaN	0.0	USD	No	Thumbnail	...	\$

5 rows × 22 columns



In [23]: df_1.head()

Out[23]:

	id	category	title	body	amenities	bathrooms	bedrooms	currency	fee	has_photo	...	price_di
0	5668640009	housing/rent/apartment	One BR 507 & 509 Esplanade	This unit is located at 507 & 509 Esplanade, R...	NaN	1.0	1.0	USD	No	Thumbnail	...	\$
1	5668639818	housing/rent/apartment	Three BR 146 Lochview Drive	This unit is located at 146 Lochview Drive, Ne...	NaN	1.5	3.0	USD	No	Thumbnail	...	\$
2	5668639686	housing/rent/apartment	Three BR 3101 Morningside Drive	This unit is located at 3101 Morningside Drive...	NaN	2.0	3.0	USD	No	Thumbnail	...	\$
3	5668639659	housing/rent/apartment	Two BR 209 Aegean Way	This unit is located at 209 Aegean Way, Vacavi...	NaN	1.0	2.0	USD	No	Thumbnail	...	\$
4	5668639374	housing/rent/apartment	One BR 4805 Marquette NE	This unit is located at 4805 Marquette NE, Alb...	NaN	1.0	1.0	USD	No	Thumbnail	...	

5 rows × 22 columns



In [20]: df.info()

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 22 columns):
#   Column                Non-Null Count  Dtype
---  ---
0   id                    10000 non-null  int64
1   category              10000 non-null  object
2   title                 10000 non-null  object
3   body                  10000 non-null  object
4   amenities             6451 non-null   object
5   bathrooms             9966 non-null   float64
6   bedrooms              9993 non-null   float64
7   currency              10000 non-null  object
8   fee                   10000 non-null  object
9   has_photo             10000 non-null  object
10  pets_allowed          5837 non-null   object
11  price                 10000 non-null  int64
12  price_display         10000 non-null  object
13  price_type            10000 non-null  object
14  square_feet           10000 non-null  int64
15  address               6673 non-null   object
16  cityname              9923 non-null   object
17  state                 9923 non-null   object
18  latitude              9990 non-null   float64
19  longitude             9990 non-null   float64
20  source                10000 non-null  object
21  time                  10000 non-null  int64
dtypes: float64(4), int64(4), object(14)
memory usage: 1.7+ MB

```

```
In [24]: df_1.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 99492 entries, 0 to 99491
Data columns (total 22 columns):
#   Column                Non-Null Count  Dtype
---  ---
0   id                    99492 non-null  int64
1   category              99492 non-null  object
2   title                 99492 non-null  object
3   body                  99492 non-null  object
4   amenities             83448 non-null  object
5   bathrooms             99429 non-null  float64
6   bedrooms              99368 non-null  float64
7   currency              99492 non-null  object
8   fee                   99492 non-null  object
9   has_photo             99492 non-null  object
10  pets_allowed          39068 non-null  object
11  price                 99491 non-null  float64
12  price_display         99491 non-null  object
13  price_type            99492 non-null  object
14  square_feet           99492 non-null  int64
15  address               7943 non-null   object
16  cityname              99190 non-null  object
17  state                 99190 non-null  object
18  latitude              99467 non-null  float64
19  longitude             99467 non-null  float64
20  source                99492 non-null  object
21  time                  99492 non-null  int64
dtypes: float64(5), int64(3), object(14)
memory usage: 16.7+ MB

```

```
In [21]: df.describe()
```

Out[21]:

	id	bathrooms	bedrooms	price	square_feet	latitude	longitude	time
count	1.000000e+04	9966.000000	9993.000000	10000.000000	10000.000000	9990.000000	9990.000000	1.000000e+04
mean	5.623396e+09	1.380544	1.744021	1486.277500	945.810500	37.695162	-94.652247	1.574891e+09
std	7.021025e+07	0.615410	0.942354	1076.507968	655.755736	5.495851	15.759805	3.762395e+06
min	5.508654e+09	1.000000	0.000000	200.000000	101.000000	21.315500	-158.022100	1.568744e+09
25%	5.509248e+09	1.000000	1.000000	949.000000	649.000000	33.679850	-101.301700	1.568781e+09
50%	5.668610e+09	1.000000	2.000000	1270.000000	802.000000	38.809800	-93.651600	1.577358e+09
75%	5.668626e+09	2.000000	2.000000	1695.000000	1100.000000	41.349800	-82.209975	1.577359e+09
max	5.668663e+09	8.500000	9.000000	52500.000000	40000.000000	61.594000	-70.191600	1.577362e+09

```
In [25]: df['category'].unique()
```

```
Out[25]: array(['housing/rent/apartment', 'housing/rent/home',  
              'housing/rent/short_term'], dtype=object)
```

```
In [28]: df['title'].unique()
```

```
Out[28]: array(['Studio apartment 2nd St NE, Uhland Terrace NE, Washington, DC 20002',  
              'Studio apartment 814 Schutte Road',  
              'Studio apartment N Scott St, 14th St N, Arlington, VA 22209', ...,  
              'Six BR 9908 Bencross Drive', 'One BR in New York NY 10069',  
              'Beautiful Lawrenceville Apartment for rent'], dtype=object)
```

```
In [27]: df['body'].unique()
```

```
Out[27]: array(['This unit is located at second St NE, Uhland Terrace NE, Washington, DC 20002, Washington, 20002, DCMonthly rental rates range from $790 - $1090We have studio units available for rent',
               'This unit is located at 814 Schutte Road, Evansville, 47712, INMonthly rental rates range from $425 - $445We have studio - 1 beds units available for rent',
               'This unit is located at N Scott St, 14th St N, Arlington, VA 22209, Arlington, 22209, VAMonthly rental rates range from $1390We have studio units available for rent',
               ...,
               'This unit is located at 9908 Bencross Drive, Potomac, 20854, MDMonthly rental rates range from $11000We have 6 beds units available for rent',
               "Monthly Rent$4,605 -to $4,790AmenitiesThe Aldyn offers some of the finest amenities amongst Upper West Side apartments. The club-style way of apartment living offers residents countless amenities right at their doorstep - the cornerstone being the 40,000 sq. feet La Palestra Athletic Club and Spa. This prestigious facility includes a 75 foot indoor pool, hot bathtub, 38 foot rock climbing wall, basketball and squash courts, personal training, bowling alley, and more!Community Amenities 75 ft indoor swimming pool and hot bathtub 38 Rock Climbing Wall Elegant Roman and Williams designed lobby & lounge Childrens indoor children's play area by KIDVILLE Pets allowed Free shuttle to Columbus Circle & 72nd/Broadway Entertainment room with gourmet catering kitchen 24hr attendant at door/concierge services On-site parking garage Private landscaped court yard by Matthews Nielson Squash court Bike Room Bowling Alley, game room and golf simulator Spa features mens & peoples locker rooms",
               "Square footage: 880 sq. feet, unit number: 15003. Nestled on a secluded hill outside of Atlanta, in Lawrenceville, Georgia, has one, 2 and 3 beds apartment homes conveniently located just off I-85, Duluth, Georgia Highways 316 and 120. Our central location makes for easy commuting to work or leisure-time destinations. Holland Park is near 2 AMC movie theaters and an IMAX theatre and just a few minutes from the Mall of Georgia, Gwinnett Mall and Discover Mills Mall. Families with children appreciate our location in the award-winning Collin's Hill High School cluster. Living in your new apartment Park Apartments, you'll be close to Lawrenceville's quaint town with boutiques, independently owned restaurants and bars, plus terrific night life and community events. On our premises, available amenities a free 24-hr state-of-the-art Fitness facilities. Enjoy a refreshing swim in 1 of our two, resort-style pools with patio areas, grills and shady arbors. Our four-legged residents will love our Bark Park, too."],
               dtype=object)
```

```
In [29]: df['amenities'].unique()
```

```
Out[29]: array([nan, 'Dishwasher,Elevator,Patio/Deck,Pool,Storage', 'Refrigerator',
               ...,
               'Cable or Satellite,Dishwasher,Fireplace,Parking,Patio/Deck,Refrigerator,Wood Floors',
               'Elevator,Gym,Parking,Patio/Deck,Pool,Storage,Tennis,View',
               'Basketball,Cable or Satellite,Doorman,Hot Tub,Internet Access,Parking,Playground,Pool,Storage,Washer Dryer'],
               dtype=object)
```

```
In [30]: df['currency'].unique()
```

```
Out[30]: array(['USD'], dtype=object)
```

```
In [31]: df['fee'].unique()
```

```
Out[31]: array(['No'], dtype=object)
```

```
In [31]: df['fee'].unique()
```

```
Out[31]: array(['No'], dtype=object)
```

```
In [32]: df['has_photo'].unique()
```

```
Out[32]: array(['Thumbnail', 'Yes', 'No'], dtype=object)
```

```
In [33]: df['pets_allowed'].unique()
```

```
Out[33]: array([nan, 'Cats,Dogs', 'Cats', 'Dogs'], dtype=object)
```

```
In [35]: df['price_display'].unique()
```

```
Out[35]: array(['$790', '$425', '$1,390', ..., '$19,500', '$25,000', '$4,790'],  
              dtype=object)
```

```
In [36]: df['price_type'].unique()
```

```
Out[36]: array(['Monthly', 'Weekly', 'Monthly|Weekly'], dtype=object)
```

```
In [37]: df['address'].unique()
```

```
Out[37]: array([nan, '814 Schutte Rd', '1717 12th Avenue', ...,  
              '5407 Abbott Place Abbott', '256 Las Entradas',  
              '9908 Bencross Dr'], dtype=object)
```

```
In [38]: df['cityname'].unique()
```

```
Out[38]: array(['Washington', 'Evansville', 'Arlington', ..., 'Saint Leonard',  
              'Chaska', 'Bella Vista'], dtype=object)
```

```
In [39]: df['state'].unique()
```

```
Out[39]: array(['DC', 'IN', 'VA', 'WA', 'NY', 'CA', 'AZ', 'NC', 'TX', 'GA', 'FL',  
              nan, 'AL', 'MD', 'CO', 'NM', 'IL', 'TN', 'AK', 'MA', 'NJ', 'OR',  
              'DE', 'PA', 'IA', 'SC', 'MN', 'MI', 'KY', 'WI', 'OH', 'CT', 'RI',  
              'NV', 'UT', 'MO', 'OK', 'NH', 'NE', 'LA', 'ND', 'AR', 'KS', 'ID',  
              'HI', 'MT', 'VT', 'SD', 'WV', 'MS', 'ME', 'WY'], dtype=object)
```

```
In [40]: df['source'].unique()
```

```
Out[40]: array(['RentLingo', 'Listanza', 'ListedBuy', 'RentDigs.com', 'GoSection8',  
              'RealRentals', 'RENTOCULAR', 'rentbits', 'Home Rentals',  
              'Real Estate Agent', 'RENTCafé', 'tenantcloud'], dtype=object)
```

```
In [ ]:
```